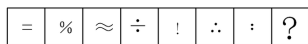


Which one of the following figures is different from the other three:



A. # B. :: C. + D. ×

Reference Solution

It is found that the number of parts of the graph in the question is: 2, 3, 2, 3, 2, 3, 2. Therefore, the graph should be a three-part graph. A, C, and D are all one-part graphs. Only option B meets the requirement. So the correct answer is B.

Owen2.5VL-72B

To solve this problem, let's analyze the sequence of symbols and try to identify any patterns or relationships that might help us determine the correct symbol for the eighth box.

Step-by-Step Analysis:

- 1st box:** $\frac{1}{2}$ (Equal sign)
- 2nd box:** $\%$ (Percent sign)
- 3rd box:** $\approx$ (Approximately equal sign)
- 4th box:** $\div$ (Division sign with dots)
- 5th box:** $!$ (Exclamation mark)
- 6th box:** $:$ (Therefore symbol)
- 7th box:** $:$ (Colon)
- 8th box:** ****Observing Patterns****

****Mathematical Symbols**:** The sequence includes several mathematical symbols such as $\frac{1}{2}$, $\%$, $\approx$, $\div$, $!$, and $:$. These symbols are commonly used in mathematical expressions and logic.

****Logical Symbols**:** The $:$ (therefore) and $:$ (colon) are often used in logical reasoning and notation.

****Progression or Relationship**:** We need to determine if there is a logical progression or relationship between these symbols.

****Possible Relationships**:**

- **Mathematical Operations**:** The symbols could represent a sequence of operations or logical steps.
- **Logical Flow**:** The $:$ (therefore) suggests a conclusion or result, and the $:$ (colon) could be used to introduce a condition or explanation.
- **Analyzing the Options**:**
 - Option A:** $\#$ (Hashtag or number sign) - This symbol is not typically used in standard mathematical or logical notation.
 - Option B:** $\because$ (Because symbol) - This symbol is used in logical notation to denote "because" or "since".
 - Option C:** $\leftarrow$ (Plus sign) - This is a common mathematical operation.
 - Option D:** $\times$ (Multiplication sign) - This is another common mathematical operation.

Conclusion:

- The $:$ (therefore) and the use of logical symbols, it seems logical to complete the sequence with a symbol that represents a reason or cause.
- The $\because$ (because) fits well in this context, as it complements the $:$ (therefore) by providing the reason or cause for the conclusion.

Final Answer:

$\because$

Image Caption

The image shows a puzzle with a sequence of symbols arranged in a horizontal row within individual square boxes. Below this sequence, there are four multiple choice options labeled A, B, C, and D, each with a distinct symbol. The puzzle sequence consists of seven square boxes arranged horizontally. Each box contains a single symbol from left to right: 1. The first box contains an equal sign '='. 2. The second box contains a percent sign '%'. 3. The third box contains an approximately equal sign '≈'. 4. The fourth box contains a division sign with a dot above and below the slash '÷'. 5. The fifth box contains an exclamation mark '!'. 6. The sixth box contains a symbol of three dots arranged in a triangular formation, with two dots on the bottom and one on top '⋮'. 7. The seventh box contains a colon ':'. The eighth box is marked with a question mark '?'. Below the sequence, there are four multiple choice options: A. A hashtag or number sign symbol '#'. B. A symbol with two vertically aligned dots with a colon between them, similar to the triangular dot pattern '⋮'. C. A plus sign '+'. D. A multiplication sign '×'. The task is likely to identify a pattern or relationship among the symbols in the sequence to determine which of the four symbols (A, B, C, or D) belongs in the eighth box marked with a question mark.

GPT-4o

To solve this problem, we need to discern the pattern or sequence in the given symbols. Looking at the sequence of symbols: $\backslash\% , \backslash\approx , \backslash\div , ! , \backslash\cdots , \backslash\!n$. Let's analyze each one: $\backslash\% =$ is a symbol for equality. $\backslash\approx$ represents approximation. $\backslash\div$ is the symbol for division. $\backslash!$ represents factorial. $\backslash\cdots$ signifies factorial. $\backslash!n$ implies "therefore." $\backslash!$ generally denotes a ratio or presence of something. To identify the pattern, consider the sequence types: $\backslash!$ The symbols seem to be mathematical or logical operators. Each symbol could represent a series or group classification in math. Now, for patterns or relationships: $\backslash!$ Equality, percentage, approximation, division: These are basic mathematical operations or relations. $\backslash\approx$ Factorial is a mathematical operation with numbers. $\backslash!$ "Therefore" and ratio both have logical or presentational meaning. Now, let's match options: A. "# - Hash, not quite fitting into the mathematical operation. B. "Because" logically follows "therefore" (quite fitting). C. "+" - Addition, fits mathematically but not particularly a logical sequence. D. "x" - Multiplication, a mathematical operator but not sequence-linked. Given that "+" means "therefore," the next logical sequence in relationships would be "because," which often appears as a subsequent explanation. Thus, the pattern aligns more with logical progression from "+" to "because." Answer: B

Doubao-1.5-Vision-Pro-32k-250115

1. Analyze the nature of the given symbols:
 - The symbols in the sequence “=, %, ~, +, !, ~,” are all special - purpose symbols often used in mathematics, logic, or general notation, not common arithmetic operation symbols like “+” and “x”.
 - Among the options: “-” Option C “+” and Option D “x” are basic arithmetic operation symbols for addition and multiplication respectively. They do not fit the pattern of non - arithmetic - operation special symbols in the given sequence.
 - Option A “#” is a general symbol often used in programming, tagging, etc., but it has no direct relation to the set of symbols related to mathematical and logical notations in the given sequence.
 - Option B “~” is a logical symbol used to mean “because” in logical and mathematical reasoning, which is in line with the nature of the given set of symbols (such as “~” which means “therefore”).
 - Conclusion:
 - The most suitable symbol to fill in the blank to maintain the pattern is “~” as it belongs to the same category of logical and mathematical notation symbols as the ones in the given sequence.
 - Answer: \boxed{B}.

Gemini-2.0-pro-exp-02-05

Detailed step-by-step derivation: **n1. **Analyze the sequence:** The given sequence consists of seven symbols followed by a question mark: '=', '%', '≈', '+', '!', '·', '×', '?'.n2. **Identify potential patterns:** ** Look for visual patterns, numerical patterns, or logical relationships between the symbols. One noticeable feature is the presence and number of dots in each symbol.n3. **Count the dots:** ** Count the number of dots (or small circles) in each symbol:n '=': 0 dots/n '≈': 2 dots/n '+' : 0 dots/n '!' : 2 dots/n '·' : 1 dot/n '×' : 3 dots/n '?' : 2 dots/n '?': Unknown.n4. **Formulate the dot sequence:** ** The sequence of the number of dots is 0, 2, 0, 1, 3, 2, 2.n5. **Look for patterns within the dot sequence:** ** The sequence 0, 2, 0, 2, 1, 3, 2 doesn't immediately suggest a simple arithmetic or geometric progression. Let's examine alternating positions:n * Odd positions (1st, 3rd, 5th, 7th): 0, 0, 1, 1, 2. This sequence appears to be non-decreasing (0, 0, 1, 2).n * Even positions (2nd, 4th, 6th, 8th): 2, 2, 3, 2, 2. This sequence starts with two 2s, then a 3. A plausible continuation would be another 3, forming the pattern (2, 2, 3, 3).n6. **Determine the required number of dots:** ** Based on the alternating pattern hypothesis, the 8th symbol (an even position) should follow the pattern (2, 2, 3, 3), meaning it should have 3 dots.n7. **Evaluate the options:** ** Check the number of dots in each of the given options:n * A. '=': 0 dots/n * B. '≈': 2 dots/n * C. '+': 0 dots/n * D. '×': 0 dots/n8. **Select the correct option:** ** Only option B (≈) has the required 3 dots.n9. **Confirm the pattern:** ** If B is correct, the complete dot sequence is 0, 2, 0, 2, 1, 3, 2, 3. The patterns for odd (0, 0, 1, 2) and even (2, 2, 3, 3) positions hold.n10Answer: The final answer is '\$boxed{B}\$'.

o3-mini

To solve the puzzle, let's analyze the sequence of symbols and identify the underlying pattern:\n/n1. **Sequence Analysis**:\n- **= (Equal Sign)**: Represents equality.\n- **% (Percent Sign)**: Represents a ratio or proportion.\n- **~ (Approximately Equal Sign)**: Indicates approximation.\n- **÷ (Division Sign with Dots)**: Represents division.\n- **! (Exclamation Mark)**: Often used to denote factorials in mathematics or emphasis in language.\n- **.** (Therefore Symbol): Indicates a conclusion.\n- **.** (Colon)**: Used to introduce a list or explanation.\n/n2. **Identifying the Pattern**:\n- The sequence transitions from mathematical symbols to logical and punctuation symbols.\n- After the \"therefore\" symbol (:), which signifies a conclusion, the colon (:) is used to introduce an explanation or reason.\n- Following this logical progression, the next symbol should logically represent \"because,\" which is denoted by the \"**\" (Because Symbol).\n- **Evaluating the Options**:\n- **Option A (#)**: Hashtag or number sign, unrelated to the pattern.\n- **Option B (:)**: Combines the colon with the \"because\" symbol, fitting the logical continuation.\n- **Option C (+)**: Plus sign, unrelated to the current logical flow.\n- **Option D (x)**: Multiplication sign, also unrelated to the current pattern.\n- **Conclusion**:\n- **Option B (:)** best fits the established pattern by introducing the \"because\" symbol, which logically follows the \"therefore\" symbol and colon.\n/n**Answer:** \nboxed{B}

Figure 18: **Solution examples generated by different models.** Reference solution and outputs generated by GPT-4o, Qwen2.5VL-72B, Gemini-2.0-pro-exp-02-05 and Doubao-1.5-Vision-Pro-32k. Additionally, the image caption and solution from LLMs (o3-mini) are also illustrated.